

EPSILON TEXPEDITION HACKATHON

TraitTrace

Privacy-First Cookie-less Personalization via Zero-Party Knowledge Graphs

FastAPI | NetworkX | Groq Llama 3.1 8B | Next.js 14 | Framer Motion | react-force-graph-2d

github.com/Aakash10032005/TRAITTRACE

Cookies are dying. Marketers are blind.

Challenge	Impact on Marketers
GDPR / CCPA regulations	Third-party data collection legally restricted
Apple App Tracking Transparency	Cross-app tracking blocked by default on iOS
Google Privacy Sandbox	Third-party cookies deprecated in Chrome
Browser tracker blockers	Safari ITP and Firefox ETP block most tracking pixels

Critical question: How do we serve personalized experiences **WITHOUT** invading user privacy?

02 // OUR SOLUTION

TraitTrace: Explicit Consent. Zero Cookies. Full Personalization.

01 INTAKE INDEX CARDS

Consumer interacts with a gamified physical-styled swipe card deck to explicitly declare visual, pricing, and lifestyle preferences. No guessing, no tracking.

02 IN-MEMORY KNOWLEDGE GRAPH

Every swipe is mapped into a session-isolated NetworkX DiGraph on the backend. Builds a real-time behavioral graph path, ephemeral by design — zero persistence.

03 GROQ LLAMA 3.1 INFERENCE

Graph-text compiler translates paths into a semantic prompt queried through llama-3.1-8b-instant. Returns hyper-personalized storefront copy in under 150ms.

04 MARKETPLACE RE-ROUTING

Tailored smartwatch recommendations rendered from the synthesized profile with direct checkout links to Amazon.in, Flipkart, and Google Shop.

End-to-End Ephemeral Data Flow

Layer	Technology	Role
Frontend UI	Next.js 14 App Router + TypeScript	Gamified swipe cards, split-screen dual interface
Animations	Framer Motion	Drag gestures, card transitions, indicator overlays
API Transport	FastAPI POST /api/interact	Receives session UUID + trait value per swipe
Graph Engine	NetworkX DiGraph (in-memory)	Session-isolated trait graph, zero persistence
LLM Inference	Groq llama-3.1-8b-instant	Graph paths to structured JSON persona, <150ms
Graph Visualizer	react-force-graph-2d	Live 2D force-directed graph, marketer dashboard
Validation	Pydantic v2 + FastAPI models	Schema enforcement on all I/O payloads

Zero-Party by Design. No Compromises.

Privacy Principle	Implementation Detail	Verifiable Outcome
No Cookies	Session ID travels in JSON body only	Browser writes zero tracking cookies
Ephemeral State	NetworkX graph lives in RAM only	Process restart = complete data wipe
Zero-Party Data	User explicitly declares all preferences	GDPR Article 6 legitimate interest basis
No Persistent DB	No SQL, no Redis, no cloud object store	Nothing to subpoena, nothing to breach
Session Isolation	UUID per page load, never persisted	Zero cross-session user linkage possible
Open Weights LLM	Groq Llama 3.1 (open-source model)	No OpenAI black-box data retention risk

A Dual-Concept Split Screen

THE MANILA FOLDER (Left Panel)	THE CONTROL ROOM (Right Panel)
Warm kraft card-stock aesthetic. The consumer's readable case file. Explicit zero-party consent UI.	
Framer Motion swipe gesture cards 4 trait-profiling index cards (Aesthetic, Value, Budget, Habit) Dynamic Groq-generated hero copy Smart product recommendation card Marketplace checkout links	

4/4 Tests Passing. 100% Critical Path Coverage.

Test Name	Validates	Result
test_graph_manager_path_creation	Node/edge creation, path string output	PASS
test_interact_endpoint_success	Groq mock returns correct persona JSON	PASS
test_interact_endpoint_groq_failure_fallback	API error gracefully falls to default	PASS
test_interact_endpoint_invalid_json_fallback	Malformed LLM output handled cleanly	PASS

Run from project root with venv active: python -m pytest

07 // INSTRUCTIONS TO RUN

Up and Running in 4 Steps

Step	Command / Action	Notes
1. API Key	Create .env in project root: GROQ_API_KEY=gsk_...	Free key at console.groq.com
2. Backend	cd backend then uvicorn main:app --reload --port 8000	Verify: localhost:8000/health
3. Frontend	cd frontend then npm install then npm run dev	Open: localhost:3000
4. Tests	python -m pytest (from project root, venv active)	4/4 pass expected

TraitTrace

**The future of personalization is explicit,
ephemeral, and earned.**

Repository: github.com/Aakash10032005/TRAITTRACE

Stack: FastAPI + NetworkX + Groq Llama 3.1 + Next.js 14

Hackathon: Epsilon TeXpedition 2024